Measure Theoretic Probability
Lesson 9
Continuity of Probability and an Introduction to Independence

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Outline

1. Setting and the Central Question
2. Continuity of Probability
3. Probabilities of $\liminf$ and $\limsup$
4. Strict Inequalities: An Example
5. Independence of Events
6. Summary and Exercises

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Recap: Limits of Sequences of Sets

Setting

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $A_1, A_2, \cdots \in \mathcal{F}$. Recall

$$\liminf_n A_n = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m, \qquad \limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m.$$

If the two sets are equal, the common set is $\lim_n A_n$; otherwise the limit does not exist.

All these sets lie in $\mathcal{F}$

Each is built from countable unions and countable intersections of sets of $\mathcal{F}$, and a σ-field is closed under both. Hence $\mathbb{P}(\liminf_n A_n)$, $\mathbb{P}(\limsup_n A_n)$ and, if the limit exists, $\mathbb{P}(\lim_n A_n)$ are all **defined**.

The Central Question

Two different limits

- $\mathbb{P}\left(\lim_n A_n\right)$: take the limit of **sets**, then apply $\mathbb{P}$.
- $\lim_n \mathbb{P}(A_n)$: apply $\mathbb{P}$ to each set, then take the limit of **numbers** in $[0, 1]$ (if it exists).

Question

When is $\mathbb{P}\left(\lim_n A_n\right) = \lim_n \mathbb{P}(A_n)$, i.e. when can the limit be moved across $\mathbb{P}$?

Analogy with continuous functions

If f is continuous and $x_n \to x$, then $f(x_n) \to f(x)$, i.e.

$$f\left(\lim_n x_n\right) = \lim_n f(x_n).$$

Regarding $\mathbb{P}$ as a function on sets, results of this type are called **continuity of probability**.

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Theorem: Continuity of Probability

Theorem

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

(a) **From below:** if $A_n \in \mathcal{F}$ and $A_n \uparrow A$ (that is, $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$), then

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

(b) **From above:** if $A_n \in \mathcal{F}$ and $A_n \downarrow A$ (that is, $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_n A_n$), then

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

Proof of (a), Step 1: Disjoint Rings

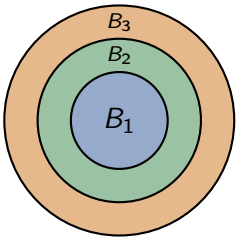
Construction

Let $A_n \uparrow A$. Put $B_1 = A_1$ and, for $n \geq 2$,

$$B_n = A_n \setminus A_{n-1} = A_n \cap A_{n-1}^c.$$

Properties

1. The B_n belong to $\mathcal{F}$ and are pairwise disjoint.
2. $\bigcup_{k=1}^{n} B_k = A_n$ for every n (by nesting).
3. $\bigcup_{n=1}^{\infty} B_n = \bigcup_n A_n = A$.



$A_1 = B_1$, $A_2 = B_1 \cup B_2$, $A_3 = B_1 \cup B_2 \cup B_3$

Proof of (a), Step 2: Countable Additivity

Computation

$$\begin{aligned}
\mathbb{P}(A) &= \mathbb{P}\Big(\bigcup_{n=1}^{\infty} B_n\Big) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) && \text{(countable additivity)} \\
&= \lim_{m\to\infty} \sum_{n=1}^{m} \mathbb{P}(B_n) && \text{(definition of an infinite series)} \\
&= \lim_{m\to\infty} \mathbb{P}\Big(\bigcup_{n=1}^{m} B_n\Big) && \text{(finite additivity)} \\
&= \lim_{m\to\infty} \mathbb{P}(A_m). && \square
\end{aligned}$$

Remark

Countable additivity is used to **enter** the series and finite additivity to **leave** it again.

Proof of (b): Reduction to (a) by Complements

Proof

Let $A_n \downarrow A$. Then $A_1^c \subseteq A_2^c \subseteq \cdots$, and by De Morgan's law

$$A^c = \Big(\bigcap_n A_n\Big)^c = \bigcup_n A_n^c.$$

So $A_n^c \uparrow A^c$, and (a) gives $\mathbb{P}(A_n^c) \to \mathbb{P}(A^c)$. By the complement rule, $\mathbb{P}(A_n^c) = 1 - \mathbb{P}(A_n)$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, hence

$$1 - \mathbb{P}(A_n) \to 1 - \mathbb{P}(A), \qquad \text{i.e.} \qquad \mathbb{P}(A_n) \to \mathbb{P}(A). \qquad \square$$

Remark: general measures

Part (a) holds for every measure μ (the proof never used $\mathbb{P}(\Omega) = 1$). Part (b) uses the complement rule, so for a general measure it needs $\mu(A_1) < \infty$; see Exercise 1.

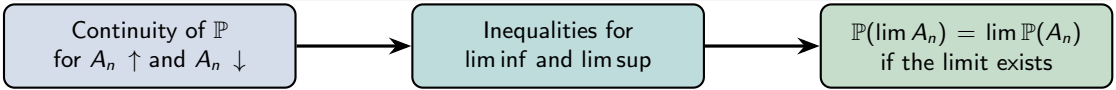
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Beyond Monotone Sequences

Motivation

Most sequences are not monotone. We want a statement of the form $\mathbb{P}(\lim_n A_n) = \lim_n \mathbb{P}(A_n)$ whenever the limit of sets **exists**, without assuming $A_n \uparrow$ or $A_n \downarrow$.



Theorem: Probabilities of lim inf and lim sup

Theorem

For **any** sequence $A_1, A_2, \cdots \in \mathcal{F}$,

$$\mathbb{P}\Big(\liminf_n A_n\Big) \;\le\; \liminf_{n\to\infty} \mathbb{P}(A_n) \;\le\; \limsup_{n\to\infty} \mathbb{P}(A_n) \;\le\; \mathbb{P}\Big(\limsup_n A_n\Big).$$

Corollary (continuity for convergent sequences of sets)

If $\lim_n A_n = A$ exists, then $\lim_n \mathbb{P}(A_n)$ exists and

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A) = \mathbb{P}\Big(\lim_{n\to\infty} A_n\Big).$$

Structure

The three inequalities are proved one at a time. The middle one is a statement about real numbers only: lim inf and lim sup of the sequence $\mathbb{P}(A_n) \in [0, 1]$.

First Inequality: $\mathbb{P}(\liminf_n A_n) \leq \liminf_n \mathbb{P}(A_n)$

Proof

Put $B_n = \bigcap_{m \geq n} A_m \in \mathcal{F}$. The sets B_n **increase**, since B_{n+1} intersects fewer sets, and $\bigcup_n B_n = \liminf_n A_n$. By continuity from below (a),

$$\mathbb{P}\Big(\liminf_n A_n\Big) = \lim_{n \to \infty} \mathbb{P}(B_n).$$

This limit exists (non-decreasing and bounded by 1), so it equals $\liminf_n \mathbb{P}(B_n)$. Since $B_n \subseteq A_n$, monotonicity gives $\mathbb{P}(B_n) \leq \mathbb{P}(A_n)$ for every n, hence

$$\liminf_n \mathbb{P}(B_n) = \liminf_n \mathbb{P}(B_n) \ \leq \ \liminf_n \mathbb{P}(A_n). \qquad \square$$

Why $\liminf$ on the right

Termwise inequalities $x_n \leq y_n$ give $\liminf x_n \leq \liminf y_n$. We cannot yet claim that $\lim \mathbb{P}(A_n)$ exists, so the right side must be a $\liminf$.

Second Inequality: A Fact About Real Sequences

Lemma

For every real sequence (a_n), $\liminf_n a_n \leq \limsup_n a_n$.

Proof

For every n, the infimum of a set is at most its supremum:

$$\inf_{m \geq n} a_m \leq \sup_{m \geq n} a_m.$$

Letting $n \to \infty$ on both sides gives the claim. □

Remark

This step has nothing to do with probability. It is applied to $a_n = \mathbb{P}(A_n)$.

Third Inequality: $\limsup_n \mathbb{P}(A_n) \leq \mathbb{P}(\limsup_n A_n)$

Proof

Put $C_n = \bigcup_{m \geq n} A_m \in \mathcal{F}$. The sets C_n **decrease**, since C_{n+1} unites fewer sets, and $\bigcap_n C_n = \limsup_n A_n$. By continuity from above (b),

$$\mathbb{P}\Big(\limsup_n A_n\Big) = \lim_{n \to \infty} \mathbb{P}(C_n).$$

Since $A_n \subseteq C_n$, monotonicity gives $\mathbb{P}(A_n) \leq \mathbb{P}(C_n)$ for every n, hence

$$\limsup_n \mathbb{P}(A_n) \;\leq\; \limsup_n \mathbb{P}(C_n) = \lim_n \mathbb{P}(C_n) = \mathbb{P}\Big(\limsup_n A_n\Big). \qquad \square$$

Proof of the Corollary: Squeezing

Proof

If $\lim_n A_n = A$ exists, then $\liminf_n A_n = \limsup_n A_n = A$. The theorem gives

$$\mathbb{P}(A) \;\leq\; \liminf_n \mathbb{P}(A_n) \;\leq\; \limsup_n \mathbb{P}(A_n) \;\leq\; \mathbb{P}(A).$$

Both ends equal $\mathbb{P}(A)$, so every term equals $\mathbb{P}(A)$. In particular $\liminf_n \mathbb{P}(A_n) = \limsup_n \mathbb{P}(A_n)$, i.e. $\lim_n \mathbb{P}(A_n)$ exists, and it equals $\mathbb{P}(A)$. □

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When Are the Inequalities Strict?

Construction

Let $A, B \in \mathcal{F}$ be **disjoint** with

$$0 < p < q < 1, \qquad p = \mathbb{P}(A), \quad q = \mathbb{P}(B).$$

Define $A_n = A$ for odd n and $A_n = B$ for even n.

The sequence of probabilities

$\mathbb{P}(A_n) = p, q, p, q, \ldots,$ hence $\liminf_n \mathbb{P}(A_n) = p$ and $\limsup_n \mathbb{P}(A_n) = q$.

The sets

Every tail intersection is $A \cap B = \emptyset$, so $\liminf_n A_n = \emptyset$. Every tail union is $A \cup B$, so $\limsup_n A_n = A \cup B$, with $\mathbb{P}(A \cup B) = p + q$ by additivity.

The Example: All Three Inequalities Are Strict

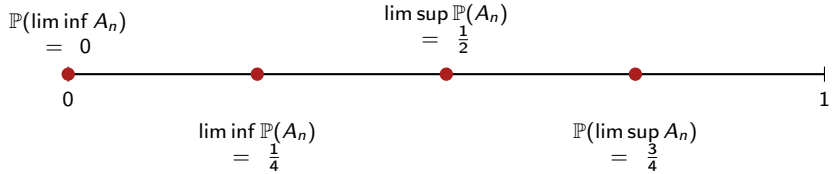
Result

$$\underbrace{\mathbb{P}\big(\liminf_n A_n\big)}_{0} < \underbrace{\liminf_n \mathbb{P}(A_n)}_{p} < \underbrace{\limsup_n \mathbb{P}(A_n)}_{q} < \underbrace{\mathbb{P}\big(\limsup_n A_n\big)}_{p+q}.$$

Here $0 < p$, $p < q$ and $q < p + q$ because $p, q > 0$.

Concrete case: two fair coin tosses

$A = \{HH\}$ and $B = \{HT, TH\}$ give $p = \frac{1}{4}$ and $q = \frac{1}{2}$.



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Independence of Two Events

Definition

Events $A, B \in \mathcal{F}$ are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Motivation via conditional probability (informal)

For $\mathbb{P}(B) > 0$, $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, and independence says $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: knowing that B occurred does not change the probability of A. The two formulations are equivalent when $\mathbb{P}(B) > 0$.

Mutual Independence of Finitely Many Events

Definition

Events $A_1, \ldots, A_n \in \mathcal{F}$ are **(mutually) independent** if for every $J \in \{2, \ldots, n\}$ and all indices $1 \leq k_1 < \cdots < k_J \leq n$,

$$\mathbb{P}\left(A_{k_1} \cap \cdots \cap A_{k_J}\right) = \prod_{i=1}^{J} \mathbb{P}(A_{k_i}).$$

Equivalent reading

The product rule must hold for **every** sub-collection with at least two events: all pairs, all triples, all quadruples, and so on.

Pairwise independence

Events are **pairwise independent** if the rule holds only for $J = 2$. Mutual independence implies pairwise independence (take $J = 2$). The converse is false, as the next example shows.

Independence of an Arbitrary Collection

Definition

Let $\{A_i : i \in I\} \subseteq \mathcal{F}$ be any collection of events (finite, countably infinite or uncountable). It is **independent** if **every finite sub-collection** is mutually independent in the sense of the previous slide.

Remarks

- The indices serve only as labels here: order plays no role.
- Independence of an infinite collection involves only finite intersections; no limits are needed.

Pairwise Independent but Not Mutually Independent

Setting: two tosses of a fair coin

$\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$. Define

Event	Description	Set
A	head on the first toss	$\{HH, HT\}$
B	head on the second toss	$\{HH, TH\}$
C	both tosses agree	$\{HH, TT\}$

Verification

- $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$.
- Every pairwise intersection is $\{HH\}$, so $\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$: **pairwise independent**.
- $A \cap B \cap C = \{HH\}$, so $\mathbb{P}(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$: **not mutually independent**.

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Summary

Key results

Result	Content
Continuity from below	$A_n \uparrow A \;\Rightarrow\; \mathbb{P}(A_n) \to \mathbb{P}(A)$
Continuity from above	$A_n \downarrow A \;\Rightarrow\; \mathbb{P}(A_n) \to \mathbb{P}(A)$
Fatou-type inequalities	$\mathbb{P}(\liminf A_n) \;\leq\; \liminf \mathbb{P}(A_n) \;\leq\; \limsup \mathbb{P}(A_n) \;\leq\; \mathbb{P}(\limsup A_n)$
Convergent sequences	$A_n \to A \;\Rightarrow\; \mathbb{P}(A_n) \to \mathbb{P}(A)$
Strictness	all three inequalities can be strict (alternating disjoint sets)
Independence	product rule for every finite sub-collection; pairwise $\not\Rightarrow$ mutual

Exercises

1. Prove continuity from below for an arbitrary measure μ. In $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ take $A_n = [n, \infty)$ and show that $A_n \downarrow \emptyset$ but $\lambda(A_n) \not\to \lambda(\emptyset)$.
2. Give disjoint sets $A \neq B$ with $\mathbb{P}(A) = \mathbb{P}(B) = p > 0$. Show $\lim_n \mathbb{P}(A_n)$ exists for the alternating sequence, although $\lim_n A_n$ does not exist.
3. If A and B are independent, prove that A and B^c are independent.
4. Show that mutual independence of $A_1, \ldots, A_n$ implies independence of every sub-collection.

Outlook

Next lesson

Start from an **independent collection** of events and form the σ-field it generates. **Question:** are the sets of the larger σ-field still independent?

Goal: the Borel–Cantelli lemmas

The first lemma concerns $\mathbb{P}(\limsup_n A_n) = \mathbb{P}(A_n \text{ i.o.})$ under a summability condition; the second requires independence, which we are now developing.